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(An Autonomous Institution)

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Gesture controlled Virtual Mouse

Abstract

The Virtual Mouse & Quantum Assistant is an AI-based system that lets users control their computer without a physical mouse or keyboard. It uses hand tracking to enable actions like moving the cursor, clicking, scrolling, and adjusting volume or brightness through gestures. It also includes Quantum, an AI assistant that supports voice and text commands, features conversational memory, integrates with language models, and provides a chat interface.

Introduction

Traditional computer interaction depends on physical devices like keyboards and mice, which can be limiting -especially for users with mobility challenges - and less natural overall. This project introduces a touchless alternative using computer vision and machine learning on standard hardware.

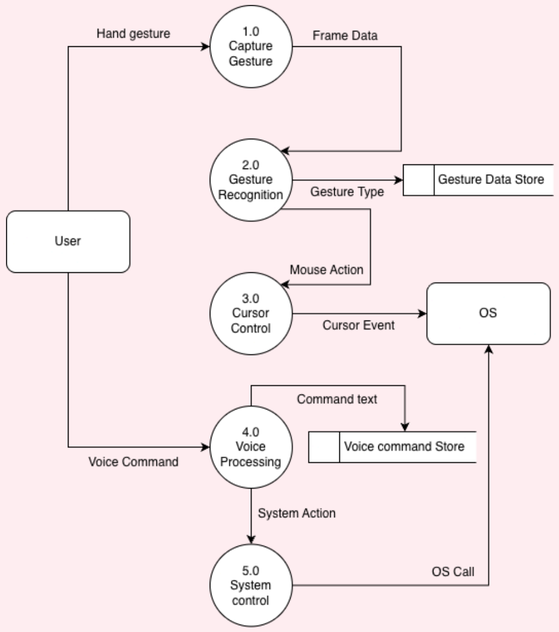
It features two modes:

- Gesture Mode, where a webcam-based virtual mouse enables cursor control, clicks, scrolling, and system adjustments through hand gestures.
- Quantum Assistant Mode, a voice and text-based chatbot that executes a wide range of system commands using fuzzy matching and an LLM-powered backend.

Objectives

- To develop a real-time, touchless mouse control system using hand gesture recognition via webcam, eliminating the need for physical input devices.
- To implement voice and text-driven AI assistant capable of dispatching 31 categories of system commands with fuzzy matching for improved accuracy.
- To integrate a cascading LLM backend with conversation memory for intelligent, context-aware responses to unrecognized commands.
- To enable custom gesture training, allowing users to define and map personal gestures to system actions, app launches, or text input.
- To provide an accessible and configurable interface through a browser-based chat UI with real-time settings for gesture sensitivity, cursor speed, and pinch threshold.

Methodology / System Design



Implementation

1.Gesture Recognition Module:
The webcam captures frames and uses MediaPipe Hands to detect 21 hand landmarks, which are then classified into gestures based on finger positions.

Figure 1 :Gesture recognition and cursor movement

2. System Control Module:
The Controller class converts detected gestures into system actions like cursor movement, clicks, and adjusting volume or brightness..

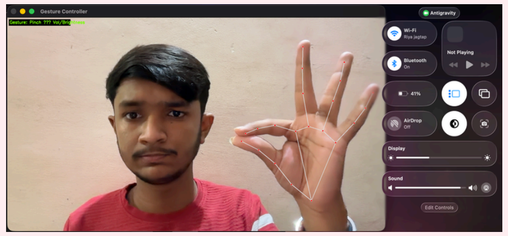


Figure 2 :Gesture-to-action mapping interface

3. Quantum Assistant - Command Dispatcher:
User input is corrected using fuzzy matching, then processed by a dispatcher that handles 31 different command categories.

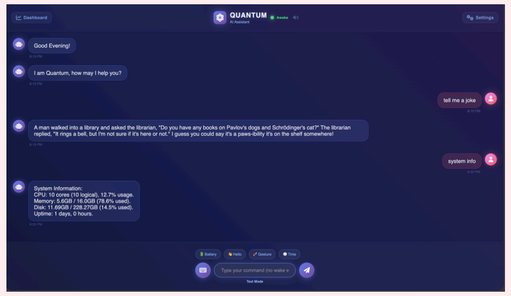


Figure 3: Quantum chat UI

4 LLM Integration & Conversation Memory:
Unrecognized commands are handled by a LangChain agent powered by models like Gemini, Groq, or Ollama.

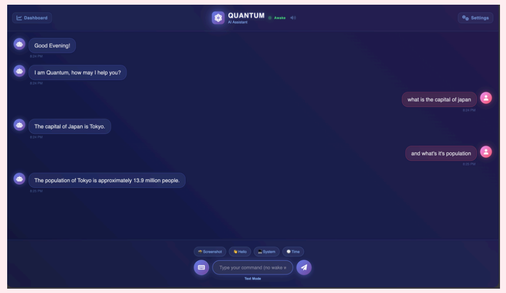


Figure 4:LLM-powered response with context

5. Live Dashboard & Settings Panel:
A browser-based UI built with Eel shows a real-time dashboard featuring gesture heatmaps, command history, and system stats.

Figure 5 : Live dashboard and settings panel

Conclusion

The Virtual Mouse & Quantum Assistant showcases an innovative approach to human-computer interaction by combining gesture-based control with an AI voice assistant. Using technologies like MediaPipe, OpenCV, and LLMs, it removes the need for physical input devices while enabling smart, context-aware commands. The system is accessible, cross-platform, and scalable, making it well-suited for modern, touchless computing environments.

References

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